handed glove. But such reasoning is refuted by the Bell/Aspect results. Now, if in response to this refutation of EPR we give up realism— as we do in standard quantum mechanics—that does nothing to lessen the stunning weirdness of long-range correlations between widely separated *random* processes; when we relinquish realism, the gloves, as in endnote 4, become "quantum gloves." Giving up realism does not, by any means, make the observed nonlocal correlations any less bizarre. It is true that if, in light of the results of EPR, Bell, and Aspect, we try to maintain realism—for example, as in Bohm's theory discussed later in the chapter —the kind of nonlocality we require to be consistent with the data seems to be more severe, involving nonlocal interactions, not just nonlocal correlations. Many physicists have resisted this option and have thus relinquished realism.

15. See, for example, Murray Gell-Mann, *The Quark and the Jaguar* (New York: Freeman, 1994), and Huw Price, *Time's Arrow and Archimedes' Point* (Oxford: Oxford University Press, 1996).

16. Special relativity forbids anything that has ever traveled slower than light speed from crossing the speed-of-light barrier. But if something has *always* been traveling faster than the speed of light, it is not strictly ruled out by special relativity. Hypothetical particles of this sort are called *tachyons.* Most physicists believe tachyons don't exist, but others enjoy tinkering with the possibility that they do. So far, though, largely because of the strange features that such a faster-than-light particle would have according to the equations of special relativity, no one has found any particular use for them—even hypothetically speaking. In modern studies, a theory that gives rise to tachyons is generally viewed as suffering from an instability.

17. The mathematically inclined reader should note that, at its core, special relativity claims that the laws of physics must be Lorentz invariant, that is, invariant under SO(3,1) coordinate transformations on Minkowski spacetime. The conclusion, then, is that quantum mechanics would be squared with special relativity if it could be formulated in a fully Lorentz-invariant manner. Now, relativistic quantum mechanics and relativistic quantum field theory have gone a long way toward this goal, but as yet there isn't full agreement regarding whether they have addressed the quantum measurement problem in a Lorentz-invariant